



Department of CSE
Related Work Report - Task 1

**“Malware Family Classification Using an Image-to-Graph GCN on
the MaleBin Dataset”**

Course Code: ICE 478/CSE475

Section: 2

Track: Track 2 - GCN (Image -> Graph)

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1. Introduction

Track 2 requires image data to be converted into graphs and classified with a graph convolutional model. Accordingly, this review focuses on recent image-to-graph and graph-based image-classification studies published from 2022 to 2025. The papers use patch graphs, superpixel region-adjacency graphs, learned positional information, multiscale superpixels, and CNN-derived graph features. They are relevant to the MaleBin project because they show different ways to define image nodes, edges, node features, and graphconvolution operations.

The reviewed papers use different datasets from MaleBin. Their reported numerical scores are therefore not treated as direct benchmarks for malware classification. The main purpose of the review is to identify graphdesign choices and a research gap that can be tested on grayscale malware byte-images.

2. Related Work Summary

Paper	Year	Dataset	Graph construction	GCN/GNN type	Representative metric
Vision GNN: An Image is Worth Graph of Nodes	2022	ImageNet-1K	Image patches as nodes; nearest-neighbor graph	ViG Grapher / graph convolution	Pyramid ViG-B: 83.7% top1
Superpixel Image Classification with GCNNs Based on Learnable Positional Embedding	2022	FashionMNIST, CIFAR-10, SVHN, CAR196, CUB200, EuroSAT	SLIC-75 superpixels; region graph + randomwalk positional encoding	ChebNet, GCN, GAT with LPE	73.09% CIFAR-10; 92.01% F-MNIST
Image Classification using GNN and Multiscale Wavelet Superpixels	2023	MNIST, FashionMNIST, CIFAR-10	WaveMesh multiscale superpixels; region adjacency graph	SplineCNN + WavePool/GraciusPool	56.89% CIFAR-10 for WaveMesh + WavePool config
Geometric Superpixel Representations for Efficient Image Classification with GNNs	2023	CIFAR-10, ImageNet-1K	Superpixels linked by region adjacency; shape + appearance features	ShapeGNN	80.44% CIFAR-10
Enhanced Image Classification with Superpixel-Derived CNN Features	2025	Four agricultural image datasets	Superpixels + CNN local/global features structured as graphs	GCN graph classifier	96.57%, 99.63%, 95.19%, 90.00% accuracy

3. Literature Review

3.1 Vision GNN: An Image is Worth Graph of Nodes (2022)

Han, Wang, Guo, Tang, and Wu proposed Vision GNN (ViG) as an alternative to treating images only as regular grids or token sequences. The method divides an image into patches and uses the patches as graph nodes. Nearest-neighbor relationships define the graph, and a Grapher module performs graph convolution to aggregate information between connected nodes. Feed-forward layers then transform node features.

The work demonstrated that graph-based vision can scale to large image-recognition tasks. The official implementation reports a top-1 ImageNet accuracy of 83.7% for Pyramid ViG-B. Its major strength is that it directly models relationships between non-local image patches. However, the architecture is considerably larger and more computationally demanding than the compact graph model used in this project.

Relation to MaleBin: the patch-node idea directly supports the use of a regular patch graph for malware byteimages. Unlike natural photographs, however, MaleBin patches do not represent semantic objects, so the graph is used as a spatial inductive bias rather than an object-relation graph.

3.2 Superpixel Image Classification with GCNNs Based on Learnable Positional Embedding (2022)

Bae et al. studied superpixel image classification with learnable positional embedding. Images are segmented into 75 SLIC superpixels and converted to graph structures. The method adds positional information initialized through random walks so that graph convolution can better distinguish different spatial arrangements. The authors evaluated ChebNet, conventional GCN, and GAT variants with the positional encoding.

The reported results include 92.01% on FashionMNIST and 73.09% on CIFAR-10 for strong IMGCN-LPE variants, with improvements over comparable graph baselines. The method is valuable because it shows that graph topology alone may be insufficient when node positions matter. Its limitation for MaleBin is that SLIC superpixels assume meaningful homogeneous regions, whereas byte-images primarily encode binary texture.

3.3 Image Classification using Graph Neural Network and Multiscale Wavelet Superpixels (2023)

Vasudevan, Bassenne, Islam, and Xing proposed WaveMesh, an adaptive wavelet-based method that produces an image-specific number and size of superpixels. The superpixels form region-adjacency graphs, and SplineCNN performs graph convolution. The paper also proposed WavePool, a pooling method that respects the multiscale wavelet hierarchy.

Experiments on MNIST, FashionMNIST, and CIFAR-10 showed that graph performance is sensitive not only to the graph representation but also to the pooling strategy. In the reported CIFAR-10 experiments, a WaveMesh configuration with WavePool reached 56.89% test accuracy. The main strength is adaptive graph resolution; the main limitation is added preprocessing complexity and sensitivity to the chosen pooling method.

3.4 Geometric Superpixel Representations for Efficient Image Classification with GNNs (2023)

Cosma, Knobel, van der Linden, Knigge, and Bekkers proposed ShapeGNN. The method constructs regionadjacency graphs by clustering pixels into superpixels and connecting adjacent regions. It enriches each superpixel with appearance, position, and shape information, including a separate patch-level GNN for shape representation.

ShapeGNN achieved 80.44% accuracy on CIFAR-10 and outperformed the authors' superpixel-based ablations. The study is important because it shows that reducing an image to a sparse graph can be efficient, but useful region geometry must be preserved. The approach is less directly transferable to malware byte-images because geometric object shape is not naturally meaningful in binary-derived textures.

3.5 Enhanced Image Classification with Superpixel-Derived CNN Features (2025)

Nguyen et al. combined superpixel graphs with features extracted by a pretrained CNN. Images are segmented into significant superpixels, local and global CNN features are extracted, and those features become node

information for a graph-classification model using GCN learning. This hybrid design allows the graph stage to operate on richer learned features rather than only simple color statistics.

The method reported accuracies of 96.57%, 99.63%, 95.19%, and 90.00% on four agricultural image datasets. Its main strength is the combination of CNN feature extraction with relational graph learning. Its limitation is the additional computational and implementation complexity introduced by preprocessing, CNN feature extraction, and graph construction.

4. Comparative Analysis

Method	Node definition	Edge strategy	Main strength	Main limitation	Relation to MaleBin
ViG	Fixed image patches	Nearest-neighbor patch graph	Simple patch-to-graph concept; scalable	Large models at high accuracy	Strong basis for patch graph
IMGCN-LPE	SLIC superpixels	Region adjacency + positional encoding	Adds spatial position to graph learning	Superpixel quality affects graph	Shows value of position/structure
WaveMesh + SplineCNN	Adaptive multiscale superpixels	Region adjacency	Adaptive graph size and pooling	More preprocessing; pooling sensitive	Useful graph-ablation lesson
ShapeGNN	Superpixels with geometry	Region adjacency	Preserves appearance, pose, and shape	Shape features less meaningful for byteimages	Highlights limits of semantic regions
CNN features + GCN	Superpixels with learned CNN features	Graph over related regions	Rich learned node features	Higher pipeline complexity	Motivates Task 3 CNNfeature GCN

5. Research Gap

The reviewed studies confirm that image-to-graph classification depends strongly on how nodes, edges, spatial information, and node features are defined. Most recent methods are designed for natural, remote-sensing, or agricultural images where superpixels correspond to visually coherent regions. MaleBin is different: its grayscale images are generated from malware bytes, so there is no guarantee that a superpixel corresponds to a meaningful malware component. This creates a clear gap for the present project: testing whether a simple, leakage-free patch graph can still provide useful relational structure for malware family classification, and determining through ablation whether the graph itself helps compared with a CNN-only representation.

A second gap concerns node features. Several graph-image methods rely on handcrafted intensity or geometry descriptors, while the 2025 hybrid study demonstrates the value of CNN-derived node features. The project therefore progresses from a first PatchGCN with simple patch statistics to an improved CNN-feature GCN, allowing a direct ablation of graph construction and feature quality on the same MaleBin data split.

6. Relation to the Proposed Work

- Graph representation: the project uses a fixed 8 x 8 patch grid, giving 64 nodes per image.
- Edges: neighboring spatial patches are connected; later ablations compare 4-neighbor and weighted 8neighbor graphs.
- Initial node features: simple local patch statistics provide a lightweight first GCN baseline.

- Improved node features: learned CNN features are used before GCN message passing in Task 3.
- Evaluation: CNN baselines and graph variants are compared on the same leakage-free split, with macro-F1 emphasized because of class imbalance.
- Interpretation: graph and XAI visualizations are treated as model-sensitivity evidence, not as semantic malware-code localization.

7. References

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